COURSE WORK: PROGRAMMING FOR DATA SCIENCE
2025.1 BATCH

MSc Data Science: Coventry University UK

***Maintain a git repository for your work for Q1 and Q2. All the codes you write must follow best practices.**

Question 1 – Enhanced OOP: University Management System (40 Marks)

Word count: 900 words

Build upon the school management system by creating a more comprehensive university system that handles multiple departments, course enrollments, and grading systems.

A. Inheritance with Multiple Classes (8 marks)

Create a base class Person and extend the inheritance hierarchy:

- Person → Student, Faculty, Staff
- Faculty → Professor, Lecturer, TA (Teaching Assistant)
- Student → UndergraduateStudent, GraduateStudent

Task: Implement the inheritance structure with proper __init__ methods and demonstrate method inheritance.

B. Advanced Student Management (8 marks)

Enhance the Student class with:

- Course enrollment system (students can enroll in multiple courses)
- GPA calculation across multiple semesters
- Academic status tracking (Good Standing, Probation, Dean's List)

Task: Create methods enroll_course(), drop_course(), calculate_gpa(), and get_academic_status().

C. Encapsulation with Validation (8 marks)

Implement data validation and security:

- Private attributes with getter/setter methods
- Input validation (e.g., GPA must be between 0.0-4.0)
- Data integrity checks (e.g., enrollment limits)

Task: Create a SecureStudentRecord class with proper encapsulation and validation.

D. Polymorphism with Method Overriding (8 marks)

Create specialized behavior for different roles:

- Override get_responsibilities() method for each person type

- Override calculate_workload() method for different faculty types
- Demonstrate polymorphism by handling different person types in lists

Task: Show how the same method call behaves differently for different classes.

E. Department and Course Management (8 marks)

Create additional classes to manage university structure:

- Department class with faculty and course lists
- Course class with enrollment limits and prerequisites
- Methods to assign faculty to courses and students to departments

Task: Implement a complete course registration system with prerequisite checking.

Question 2 –Data Analysis: E-commerce Data Analysis (40 Marks)

Word count: 1300 words

Scrape and analyze data from multiple sources to understand patterns in online retail pricing, product ratings, and inventory management.

A. Multi-Source Data Collection (10 marks)

Web Scraping Enhancement:

- Scrape book data from http://books.toscrape.com (practice site with no restrictions)
- Collect product information from simple e-commerce demo sites
- Gather structured data from RSS feeds (XML parsing)
- Extract numerical and categorical data for analysis

Requirements:

- Use BeautifulSoup for HTML parsing and requests for data fetching
- Implement proper error handling and retry logic
- Add delays between requests (good practice)
- Save data in structured format (CSV/JSON)
- Handle pagination and multiple pages

B. Data Cleaning and Preprocessing (8 marks)

Enhanced Data Processing:

- Text preprocessing (remove special characters, normalize text)

- Handle missing data and duplicates
- Extract hashtags, mentions, and keywords
- Date/time standardization across different sources
- Data validation and quality checks

Task: Create a comprehensive data cleaning pipeline that handles messy real-world data.

C. Advanced Statistical Analysis (10 marks)

Numerical and Categorical Data Analysis:

- Price distribution analysis across different categories
- Rating patterns and statistical summaries
- Correlation analysis between price, rating, and availability
- Category popularity analysis (frequency distributions)
- Comparative analysis between different data sources

Requirements:

- Calculate descriptive statistics (mean, median, mode, standard deviation)
- Identify outliers in pricing and ratings
- Perform correlation analysis between numerical variables
- Create frequency distributions for categorical data
- Use hypothesis testing to compare groups (e.g., fiction vs non-fiction prices)

D. Visualizations (8 marks)

Statistical and Comparative Plots:

- Price distribution histograms and box plots by category
- Rating vs price scatter plots with trend lines
- Bar charts showing category popularity and average ratings
- Interactive plots using Plotly for filtering and exploration
- Comparative analysis plots between different data sources

Task: Create a comprehensive analysis report with multiple visualizations that reveal insights about pricing patterns, rating distributions, and category trends.

E. Predictive Analysis (4 marks)

Basic Statistical Modeling:

- Use simple linear regression to predict book prices based on ratings
- Identify patterns in category pricing
- Create a basic recommendation system using statistical similarity
- Analyze trends in stock availability vs pricing

Question 3 – Data Ethics: AI Ethics in Healthcare Data (20 Marks)

Word count: 1000 words

Examine the ethical challenges of using AI and data science in healthcare, focusing on bias, privacy, and algorithmic fairness.

A. Healthcare Data Privacy Challenges (5 marks)

Beyond GDPR Compliance:

- Analyze HIPAA requirements for healthcare data in the US
- Compare healthcare privacy laws across different countries
- Discuss challenges of anonymizing medical data
- Address the balance between research benefits and privacy protection

Task: Explain specific healthcare data challenges that go beyond general GDPR compliance.

B. Algorithmic Bias in Medical AI (5 marks)

Bias Detection and Mitigation:

- Identify sources of bias in medical datasets (demographic, geographic, socioeconomic)
- Explain how biased medical AI can perpetuate health disparities
- Discuss fairness metrics for healthcare algorithms
- Propose methods to detect and reduce bias in medical AI systems

Task: Provide real-world examples of AI bias in healthcare and suggest mitigation strategies.

C. Ethical Decision-Making Framework (5 marks)

Practical Ethics Implementation:

- Develop an ethical checklist for healthcare data science projects
- Discuss informed consent in the age of big data and AI
- Address the "right to explanation" for medical AI decisions
- Consider the ethical implications of predictive healthcare models

Task: Create a practical framework that healthcare data scientists can use to evaluate their projects.

D. Stakeholder Impact Analysis (5 marks)

Multi-Perspective Evaluation:

- Analyze impact on patients, healthcare providers, and researchers
- Discuss the role of data scientists in ethical healthcare AI
- Consider economic and social implications of healthcare AI
- Address global health equity issues in AI development

Task: Demonstrate understanding of how healthcare AI affects different stakeholders and propose balanced solutions.

Technical Requirements

Git Repository Structure

```
project/
├── question1_university_system/
│   ├── main.py
│   ├── person.py
│   ├── student.py
│   ├── faculty.py
│   └── department.py
├── question2_social_media_analysis/
│   ├── data_collection/
│   ├── data_processing/
│   ├── analysis/
│   └── visualizations/
├── question3_ethics_report/
│   └── healthcare_ethics_report.md
├── requirements.txt
└── README.md
```

Code Quality Expectations

- Clear, readable code with meaningful variable names
- Proper commenting and docstrings
- Error handling for common scenarios
- Modular code organization
- Requirements.txt with all dependencies

Enhanced Features (Bonus Points)

- Unit tests for key functions
- Interactive Jupyter notebook presentation
- Creative data visualization approaches

Submission Guidelines

1. Git Repository: Regular commits showing development progress
2. Documentation: README explaining how to run each component
3. Code Comments: Explain complex logic and design decisions
4. **Technical Report Requirements**: Your submission must include a professionally formatted PDF technical report covering all three questions to demonstrate technical competency and analytical thinking. The report structure includes: Executive Summary with overview and key findings,
    - Q1 covering architecture and design decisions
    - Q2 Data Analysis detailing methodology, EDA findings with graphs, and business insights
    - Q3 Ethics, presenting framework development and recommendations; Technical Implementation
    - discussing code quality, deployment and testing approaches
    - Reflection (200 words) on learning outcomes and future work.
    - The report must use clear section headings, include relevant diagrams/charts/code screenshots, follow academic referencing standards, and include appendices for additional materials.